

A Three-Tier Knowledge Graph RAG for Vietnamese Legal Question Answering

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Abstract. Accurate legal question answering demands navigating hierarchical document structures, following cross-references, and multi-hop reasoning, capabilities that standard Retrieval-Augmented Generation (RAG) systems lack. Vietnamese legal text compounds this challenge through tonal ambiguity, domain-specific abbreviations, and rigid structural hierarchies. This paper presents a three-tier knowledge graph RAG architecture that integrates a domain ontology, a knowledge graph, and a document tree forest. Three retrieval tiers, LLM-guided tree traversal, dual-level semantic search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion with knowledge graph expansion, operate over these representations. Evaluation on 530 expert-annotated questions across enterprise law, traffic safety law, and postgraduate education regulation shows that the proposed system retrieves at least one correct article in the top 10 results for 92.2% of queries, outperforming the strongest baseline by 7.4%. These results hold across all three domains without domain-specific tuning. To the best of the authors' knowledge, no prior work combines ontology-enhanced knowledge graphs with multi-tier retrieval for Vietnamese legal question answering.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval, Multi-Domain

1. Introduction

Accurate access to legal information is critical for citizens, businesses, and legal practitioners navigating regulatory compliance. Legal forums and social networks in Vietnam are flooded with questions on business formation, traffic penalties, and regulatory compliance, yet answers are often fragmented, contradictory, or cite repealed provisions.

Legal documents exhibit complex hierarchical structures: a single law organizes content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope. They also contain pervasive cross-references: answering “If a member borrows money to meet charter capital and later withdraws it, what are the penalties?” requires chaining Article 47 (member obligations) to Article 16, Clause 5 (prohibited acts) across different chapters. Vietnamese compounds this challenge through tonal

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ambiguity, domain-specific abbreviations (*HDQT*, *CTCP*), and a rigid legal hierarchy (Constitution, Codes, Decrees, Circulars).

Standard RAG [1] retrieves by vector similarity [2], an approach effective in many domains [3] but inadequate for legal text: embeddings blur precise legal meanings (semantic gap), flat retrieval ignores document hierarchy (no structural awareness), and dense search cannot chain cross-references (no multi-hop reasoning). LightRAG [4] and GraphRAG [5] model entity relationships but disregard document hierarchy; RAPTOR [6] builds summary trees via recursive clustering but relies on embedding-only retrieval; PageIndex [7] exploits structure but lacks semantic reasoning; legal ontology systems [8,9] formalize domain knowledge but remain disconnected from retrieval pipelines. To address these limitations, a three-tier knowledge graph RAG is proposed that unifies structural navigation, semantic retrieval, and rank fusion: Tier 1 performs LLM-guided tree traversal, Tier 2 applies dual-level semantic retrieval with intent-aware Personalized PageRank, and Tier 3 merges outputs through Reciprocal Rank Fusion with knowledge graph expansion.

This paper makes three contributions. First, a data engineering pipeline is developed that parses Vietnamese legal documents into hierarchical trees, constructs a domain ontology with 14 entity types, and builds a knowledge graph through semi-automatic LLM extraction with expert refinement (5,936 entities, 5,963 relations). Second, a three-tier RAG architecture is designed that unifies structural tree traversal, semantic graph retrieval, and reciprocal rank fusion into a single pipeline. Third, a multi-intent Personalized PageRank algorithm is introduced that adapts edge weights to detected query intent, allowing the graph to emphasize different relation types for penalty, procedural, or definitional questions. Experiments on 530 questions across three domains show that the system reaches 92.2% top-10 hit rate on legal QA, surpassing PageIndex by 7.4% and RAPTOR by 51.2%, and 88.5% on education regulation, exceeding PageIndex by 13.9%.

2. Related Work

Neural-based information retrieval. Dense retrieval encodes queries and passages into shared embedding spaces for similarity search. Sentence-BERT [2] introduced bi-encoder sentence embeddings that enable efficient retrieval, though compressing all semantics into a fixed-size vector limits fine-grained interaction between query and passage tokens. DPR [10] optimized this paradigm for open-domain QA with contrastive training on question-passage pairs. Advanced RAG designs incorporate self-reflection [11], corrective retrieval, and hypothetical document generation [12]; Fan et al. [3] survey this progression. Domain-adapted transformers such as LEGAL-BERT [17] pre-train on legal corpora (EU legislation, court cases, contracts) to capture domain-specific phrasing, but operate on flat passage representations without exploiting document hierarchy. For Vietnamese, PhoBERT [19] and VnCoreNLP [20] provide foundational NLP models. Ngo et al. [21] combined BM25 with BERT [22] and legal-domain heuristics for the ALQAC 2021 Vietnamese legal QA competition, achieving competitive performance on single-document tasks but without graph-based reasoning for cross-document references. A common limitation across neural approaches is the loss of structural awareness critical for navigating legal document hierarchies.

Symbolic and ontology-based retrieval. Knowledge graphs enable structured reasoning over domain knowledge. GraphRAG [5] partitions entity graphs into communities and generates LLM summaries for query-focused retrieval, enabling multi-hop reasoning but treating documents as bags of entities without preserving hierarchy. LightRAG [4] combines vector search with knowledge graph traversal in a dual-level architecture and reports faster indexing than GraphRAG; however, in our experiments on Vietnamese legal text, entity extraction quality was low, 74% of extracted entities appeared only once, limiting graph utility for structured retrieval. Legal-Onto [8] combines the Rela-model [18] with conceptual graphs to formalize Vietnamese Land Law, enabling rule-based legal reasoning but relying on manual inference rules not designed for retrieval at scale. Surveys [23,24] confirm that bridging formal legal knowledge with scalable retrieval remains open.

LLM-based approaches. RAG [1] grounds LLM generation in retrieved evidence, reducing hallucination but remaining bottlenecked by retrieval quality, when the retriever misses relevant documents, generation accuracy degrades accordingly. RAPTOR [6] builds summary trees via recursive clustering and LLM summarization, capturing multi-granularity themes; however, it discards original document hierarchy and relies on embedding-only retrieval at query time, achieving only 41.0% top-10 hit rate on our structured legal benchmark. PageIndex [7] uses LLM-guided tree traversal for hierarchical document search, achieving high precision when the model selects the correct branch, but missing content outside the traversal path; it also lacks knowledge graph integration for cross-document reasoning.

Positioning. No existing system combines ontological knowledge, graph-based retrieval, and structural tree navigation for Vietnamese legal text. Neural approaches achieve broad coverage but lack structure; symbolic systems enable reasoning but lack scalability; LLM-based systems provide language understanding but hallucinate without domain grounding. Our system integrates all three capabilities through a three-tier architecture with intent-aware reasoning and rank fusion.

3. Methodology

3.1. Problem Formulation

Given a legal corpus of m documents, each with a rigid hierarchy (Document $\rightarrow$ Chapter $\rightarrow$ Section $\rightarrow$ Article $\rightarrow$ Clause $\rightarrow$ Point), and a user query q , the task is to retrieve a ranked list of the k most relevant articles. The retrieval must operate at *article-level* granularity, as Vietnamese legal practice requires citing specific articles rather than text passages. This differs from standard passage retrieval in two aspects: (1) the search space is a forest of hierarchical trees rather than a flat document collection, and (2) answering a query may require multi-hop reasoning across articles in different chapters or documents linked by cross-references.

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3.2. System Overview

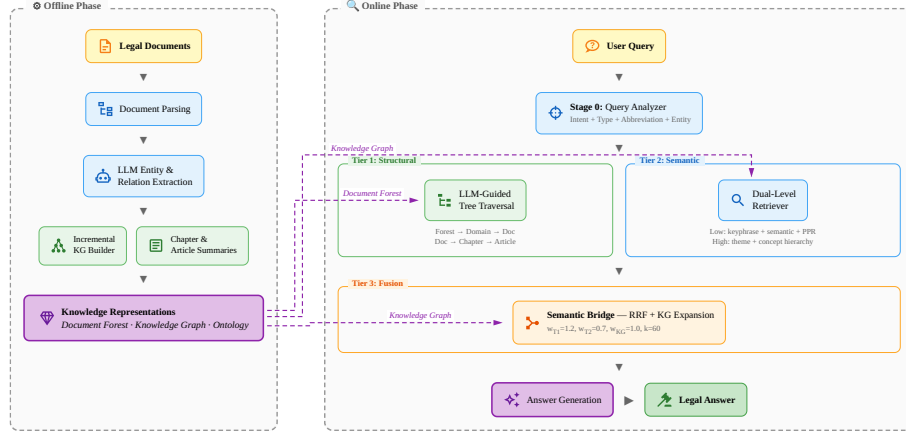


Figure 1. Architecture of the proposed three-tier retrieval system. Offline phase (left): knowledge model construction from legal documents. Online phase (right): query processing through Tier 1 (structural tree traversal), Tier 2 (dual-level semantic retrieval), and Tier 3 (RRF fusion with KG expansion).

Figure 1 illustrates the architecture, which separates into an offline phase (left) and an online phase (right). The offline phase parses legal documents into hierarchical trees, extracts entities and relations into a knowledge graph, and generates summaries for each structural level. The online phase processes each query through four stages: query analysis (Stage 0), LLM-guided tree traversal (Tier 1), dual-level semantic retrieval over the knowledge graph (Tier 2), and fusion with knowledge graph expansion (Tier 3).

3.3. Data Engineering and Knowledge Construction

The offline phase builds three knowledge representations from the raw corpus, following knowledge engineering practice [9]: (1) a **document forest** where each document is a rooted tree with six hierarchical levels and cross-reference edges link articles across documents; (2) a **knowledge graph** with 14 entity types and 56 relation types, constructed via LLM extraction with expert-in-the-loop refinement (~ 130 manually added cross-document relations); and (3) **summaries** at chapter and article level for LLM-guided branch selection. Implementation details are described in Section 4.

3.4. Hybrid Retrieval Architecture

3.4.1. Stage 0: Query Analysis

The query analyzer performs: (1) intent classification into six types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex with LLM fallback; (2) abbreviation expansion (e.g., *TNHH* $\rightarrow$ *trách nhiệm hữu hạn*); and (3) entity extraction of article references, law references, and legal keywords to seed retrieval. Each

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query intent amplifies its most relevant relation types during graph traversal (weight 1.0 for primary relations, 0.5 for others).

3.4.2. Tier 1: LLM-Guided Tree Traversal

Tier 1 navigates the document forest using LLM reasoning in three loops. Loop 0 (Forest→Domain→Document) matches the query to a domain group using pre-computed keywords, then selects the top 5 documents. Loop 1 (Document→Chapter) selects the top 8 chapters using chapter summaries. Loop 2 (Chapter→Article) identifies the top 7 articles from article summaries. General Provisions chapters are automatically included. Each loop produces a reasoning trace for interpretability.

3.4.3. Tier 2: Dual-Level Semantic Retrieval

Tier 2 performs hybrid semantic search over the knowledge graph at two levels. The *low level* (entity-specific) combines keyphrase matching, query-entity cosine similarity, and intent-aware Personalized PageRank (PPR). The *high level* (conceptual) identifies legal themes and traverses the ontology class hierarchy, surfacing related articles even without direct entity matches.

The PPR component computes query-specific node importance on the knowledge graph [16]:

$$\text{PPR}(v) = (1 - \alpha) \sum_{u \in N(v)} \frac{w(u, v) \cdot \text{PPR}(u)}{d(u)} + \alpha \cdot p(v) \quad (1)$$

where $\alpha = 0.15$ is the teleport probability, $w(u, v)$ is the edge weight adjusted by query intent, $d(u)$ is the weighted out-degree, and $p(v)$ is the personalization vector seeded from matched entities.

3.4.4. Tier 3: Semantic Bridge (RRF Fusion)

Tier 3 merges results from Tier 1 and Tier 2 using Reciprocal Rank Fusion [25]:

$$\text{RRF}(d) = \sum_{s \in \{T1, T2, KG\}} w_s \cdot \frac{1}{k + \text{rank}_s(d)} \quad (2)$$

where $k = 60$ and source weights are $w_{T1} = 1.2$, $w_{T2} = 0.7$, $w_{KG} = 1.0$. Tree traversal receives the highest weight as it provides the most reliable structural signal; the lower weights for Tier 2 and KG allow them to compensate when the LLM selects suboptimal chapters.

After fusion, results are expanded through the knowledge graph’s relation edges. Strong relations (*references*, *requires*, *depends_on*) are expanded both cross-chapter (up to 5 articles) and same-chapter (up to 2 articles), with a semantic relevance filter (cosine similarity ≥ 0.4) to suppress noise. Adjacent articles (± 2) from each result are added with decay factor 0.7, as correct answers often lie in neighboring articles. The final ranked list is passed to an LLM for answer generation.

4. Implementation

The system is implemented in Python 3.11 with the following technical choices.

Document parsing. Legal texts are parsed with custom regex parsers that detect Vietnamese hierarchy markers (“Chuong”, “Muc”, “Dieu”, “Khoan”, “Diem”) and build six-level trees. For education regulations, four scanned PDFs (118 pages total) are converted to images at 300 DPI via pdf2image² and processed via LLM Vision OCR (Gemini 2.5 Flash³); one text-based PDF is extracted with pdfplumber⁴. A hierarchy parser then converts raw text into structured JSON trees. Cross-references between documents are extracted via pattern matching on article citations (e.g., “Dieu 17 Luật Doanh nghiệp”).

Summary generation. Chapter-level and article-level summaries are generated via LLM (Claude 3.5 Haiku⁵), producing keyword-rich descriptions that Tier 1 tree traversal uses for branch selection without reading full article text.

Knowledge graph construction. Entity extraction and relation identification use Claude 3.5 Haiku via the Anthropic API, with one LLM call per article. Vietnamese slug normalization removes diacritics and collapses tonal variations to produce canonical entity identifiers. A multi-step entity resolver (exact match, synonym resolution, abbreviation expansion) merges equivalent entities across documents. The domain ontology defines 14 entity types using OWL classes with a 4-level hierarchy and Vietnamese label mappings.

Online query processing. The query analyzer classifies intent into six types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex patterns with LLM fallback. Tree traversal and article selection use Claude Sonnet 4 via the Anthropic API. Sentence embeddings are computed with paraphrase-multilingual-MiniLM-L12-v2⁶, a 384-dimensional multilingual model from the Sentence-Transformers library [2]. Vector similarity search uses FAISS⁷. Personalized PageRank is computed iteratively (max 50 iterations, tolerance 10^{-6} , damping $\alpha = 0.15$). All LLM responses are cached in SQLite for reproducibility.

5. Experiments

5.1. Data Collection

Legal corpus. The evaluation corpus consists of 18 Vietnamese legal and regulatory documents spanning three domains. The enterprise law domain includes Law 59/2020/QH14 (Enterprise Law) and nine associated decrees, totaling 77 chapters and 840 articles. The traffic safety law domain covers Law 36/2024/QH15 and two penalty decrees. The education regulation domain comprises five postgraduate documents from UIT and VNUHCM (109 articles, 376 clauses, 213 points); four of these were scanned PDFs requiring LLM

²<https://github.com/Belval/pdf2image>

³<https://deepmind.google/technologies/gemini/flash/>

⁴<https://github.com/jsvine/pdfplumber>

⁵<https://www.anthropic.com/claude>

⁶<https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>

⁷<https://github.com/facebookresearch/faiss>

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Vision OCR (Gemini 2.5 Flash, 118 pages) and one was extracted with pdfplumber. In total, the corpus contains 949 articles. Legal texts were collected from ThuVienPhapLuat⁸, while education regulations were obtained from the UIT Graduate Training Portal⁹.

Evaluation questions. 530 questions were collected from two sources: legal domain questions (400) were crawled from Vietnamese Facebook legal advice groups, and education domain questions (130) were collected from UIT student forums¹⁰. A domain expert then labeled each question with 1–8 relevant articles (average 2.3), ensuring ground-truth annotations reflect legal accuracy rather than surface-level keyword matching.

5.2. Experimental Setup

Table 1 summarizes the constructed knowledge graph.

Table 1. Knowledge graph statistics.

Entity Type	Count	Relation Type	Count	Structure	Count
Total Entities	5,936	Total Relations	5,963	Tree Nodes	9,226
Action	1,903	requires	1,377	Documents	13
Legal Term	1,632	applies_to	810	Chapters	77
Organization	500	has_authority	506	Articles	840
Legal Reference	451	references	434	Cross-refs	168
Other (9 types)	1,450	Other (52 types)	2,506		

Metrics. Since each query has expert-annotated relevant articles, we adopt standard retrieval metrics [16] that enable objective, reproducible evaluation against known ground truth, rather than LLM-as-judge scores used in open-ended generation benchmarks. Top- k hit rate = $\frac{1}{|Q|} \sum_q \mathbf{1}[\exists a \in R_q : \text{rank}(a) \leq k]$ measures the fraction of queries for which at least one relevant article appears in the top- k results, where R_q denotes the set of relevant articles for query q . Mean Reciprocal Rank is defined as $\text{MRR} = \frac{1}{|Q|} \sum_q (\min_{a \in R_q} \text{rank}(a))^{-1}$, averaging the reciprocal rank of the first relevant result. Recall@ k , Precision@ k , and NDCG@ k [16] are also computed. Results are reported at $k \in \{1, 3, 5, 10, 20, 30\}$, with top-10 hit rate and MRR as the primary metrics. Top-10 hit rate is prioritized because in a RAG pipeline the LLM provides an additional validation layer over retrieved context, at least one correct article among ten candidates suffices for accurate answer generation while limiting noise that could induce hallucination. MRR complements top-10 hit rate by rewarding higher placement of relevant articles, reducing irrelevant context that the LLM must filter.

5.3. Baselines

Traditional: BM25 [15] and TF-IDF [16] with syllable-level tokenization. RAG systems: NaiveRAG [2] (dense retrieval, no reranking); LightRAG [4] (hybrid KG retrieval); RAPTOR [6] (collapsed-tree with LLM summaries); PageIndex [7] (two-step hierarchical tree search). All systems use Sonnet 4 for LLM calls and retrieve from the full corpus.

⁸<https://thuvienphapluat.vn>

⁹<https://sdh.uit.edu.vn>

¹⁰<https://forum.uit.edu.vn>

5.4. Main Results

The proposed system achieves 92.2% top-10 hit rate (MRR 0.603), outperforming PageIndex by 7.4% and BM25 by 44.7% (Table 2). PageIndex benefits from tree navigation but misses cross-reference articles outside the traversal path. RAG baselines (NaiveRAG 32.8%, LightRAG 39.8%, RAPTOR 41.0%) underperform BM25 (47.5%): general-purpose embeddings lack Vietnamese legal vocabulary, generic KG extraction introduces noise, and RAPTOR’s summaries lose article-level granularity. Our system overcomes these limitations via LLM-guided navigation and domain-specific knowledge graph.

Table 2. Legal domain hit rate (%) on 400 questions. Best in bold, second underlined.

Method	Top 5	Top 10	Top 20	Top 30	MRR
<i>Traditional retrieval</i>					
BM25	30.8	47.5	66.2	74.8	0.200
TF-IDF	29.2	45.8	64.5	74.8	0.207
Keyword	33.5	48.5	63.2	72.8	0.222
<i>RAG systems</i>					
NaiveRAG [2]	22.8	32.8	51.8	60.5	0.146
LightRAG [4]	25.5	39.8	47.8	49.2	0.173
RAPTOR [6]	25.2	41.0	54.5	62.3	0.170
PageIndex [7]	<u>76.2</u>	<u>84.8</u>	<u>90.2</u>	<u>91.0</u>	<u>0.559</u>
3-Tier (ours)	74.5	92.2	94.8	95.0	0.603

5.5. Multi-Domain Performance

Table 3 reports per-domain results, including education as a generalization test. The system outperforms all baselines across all three domains. Traffic law achieves the highest score (98.0%) due to direct violation-to-penalty mappings. Enterprise law is harder (86.5%) due to cross-decree references. Education (88.5%) shows the largest margin over PageIndex (+13.9%), where semantic retrieval captures terminology that tree navigation misses. No domain-specific tuning was applied.

Table 3. Per-domain top-10 hit rate (%) across three domains.

Method	Enterprise (197)	Traffic (203)	Education (130)
NaiveRAG	29.9	35.5	50.0
LightRAG	20.3	58.6	53.1
RAPTOR	18.5	63.5	53.8
BM25	31.5	63.1	70.0
TF-IDF	39.1	52.2	68.5
Keyword	20.3	75.9	56.2
PageIndex	<u>76.6</u>	<u>92.6</u>	<u>74.6</u>
3-Tier (ours)	86.5	98.0	88.5
<i>Ours vs. PageIndex</i>	+9.9	+5.4	+13.9

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5.6. Analysis

Per-category top-10 hit rate ranges from 0% (household business, 4 queries outside the corpus) to 100% (speed penalties). A multi-hop example: “If a member borrows money to meet charter capital and later withdraws it, what are the penalties?” Tier 1 reaches Art. 47 (member obligations); PPR walks *requires* edges to Art. 16 Cl. 5 (prohibited acts) in a different chapter; RRF places both in the top 5. No baseline retrieves this at top 30. Of the 31 missed queries (7.8%), 27 involve enterprise law: cross-decree dependencies (6), out-of-corpus articles (4), and multi-article queries (>2 expected, 14 of 31).

6. Conclusion

This paper presented a three-tier knowledge graph RAG architecture for Vietnamese legal question answering, combining structural tree traversal, semantic graph search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion. Experiments across three domains demonstrate consistent superiority over all baselines without domain-specific tuning. Limitations include Tier 1 latency ($\sim 7s$ per query due to multiple LLM calls) and lower performance on enterprise queries requiring multi-document cross-references. Future work includes extension to additional regulatory domains and multi-turn conversational support.

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